

1. What is the present status of irrigation development in Nepal? What are the factors to be considered in accelerating the development of round year irrigation in Nepal?

Answer:

Present status of irrigation development in Nepal:

- The history of irrigation in Nepal before 1922 indicate that irrigation was developed, operated and maintained by farmers known as Farmers Managed Irrigation System (FMIS).
- Out of total potential irrigable area, 76% lies in Terai region of Nepal.
- Based on the Irrigation Master Plan 2019 (Updated 2024) published by the Department of Water Resources and Irrigation (DWRI), Out of total geographical area of Nepal i.e., 14.7 million ha, 2.97 million ha is the agricultural land. Master Plan identifies about 2.5 million hectares as potentially irrigable. By 2024, irrigation infrastructure had reached approximately 1.5 million hectares, but only about one-third of this area receives reliable year-round irrigation.
- From 1922 to 1957, Government made effort to develop irrigation infrastructures in Nepal. Chandra Nahar, Juddha Nahar, Jagadispur (Banganga), Phewa Dam are few examples of the projects developed during that period.
- The minor irrigation program was introduced in the second three- years development plan to provide low-cost irrigation facilities to farmers within short period of time such as small wells, irrigation tanks, reservoir, pumps etc.
- Until mid-1980s, irrigation development by the government focused largely on the construction of physical infrastructures of canals and structures, and little attention was given to the effective management of the completed systems.
- The implementation of number of management-oriented projects were carried out such as USAI-funded Irrigation Management Transfer Project (IMTP) in 1985, Irrigation line of Credit (ILC) in 1988 financed by the World Bank, Irrigation Sector Project (ISP) IN 1988 financed by World Bank and Asian Development Bank (ADB) and Irrigation Sector Support Project (ISSP) in 1989 under ci-financing of UNDP.
- Promulgation of Irrigation Regulations (IR) in 1989 provided emphasis with water users in all phases of irrigation projects- planning, construction, operation and maintenance.
- Promulgation of Water Resources Act and irrigation Policy in 1992 with the clear vision of irrigation development, which was later amended in 1997.
- Now, Irrigation Policy 2004 is in practice.
- Similarly, Irrigation Master Plan 1990, Agriculture Perspective plan 1995, Water Resources Strategy 2002 and National Water Plan 2005 guides irrigation development in Nepal.
- Sikta Irrigation Project in Banke, Babai Irrigation Project in Bardiya, Sunkoshi Marine Diversion Project in Kailali are some of the few under construction irrigation project in Nepal.
- Irrigation Projects like Rani Jamara Kulariya Irrigation Project, Bheri-Babai Diversion Multipurpose project are ongoing irrigation projects in Nepal.
- Sunsari-Morang Irrigation system, Kankai Irrigation project, Bagmati irrigation project etc. are the major completed irrigation schemes of Nepal.

The factors that be considered for accelerating the development of year-round irrigation in Nepal are as follows:

i. Reliable Water Sources:

- Water sources that can supply water all-round the year is tapped for irrigation system.
- Perennial river sources are generally preferred for tapping the required amount of water for irrigation purpose.

- ii. **Efficient Water Management**
 - Different irrigation techniques can be used for efficient water management such as sprinkler irrigation and drip irrigation.
- iii. **Watershed Conservation**
 - Catchment areas are protected through afforestation and soil conservation which reduces erosion and sedimentation in reservoirs and canals.
- iv. **Energy Availability**
 - The electricity or solar power for groundwater pumping and lift irrigation.
 - Promote renewable energy-powered irrigation systems.
- v. **Farmer Participation and Water User Associations (WUAs)**
 - User's committee strengthen community participation in planning, operation, and maintenance.
 - It also enhances the capacity of Water User Associations for sustainable management.
- vi. **Financial Support and Investment**
 - Government budget allocation for irrigation must be increased and provide subsidies and affordable credit for irrigation equipment.
 - Private sector and donor investment must also be increased.
- vii. **Cropping Pattern and Agricultural Practices**
 - Crops suited to available water must be promoted.
 - Multiple crops cultivation and high-value agriculture of superior crops should be encouraged.

2. **Name the method of applying water in irrigation fields. How do you define small scale irrigation project in context of Nepal? Write down in the brief its scope in Nepal.**

Answer:

Irrigation is an **artificial application of water to agricultural land** in the required quantity and at the required time to supplement natural rainfall for optimum crop growth and yield.

There are several methods of irrigation adopted based on the suitability of local condition such as soil characteristics, topographic conditions, crop rotation, available water flow and others. Few methods of applying water in irrigation fields are briefly described below:

i. Surface irrigation:

- Surface irrigation is the most common type of the irrigation where the water is applied on the field in varied quantities and at different times.
- In this method, stream of water is diverted from the head of a field into the furrows or borders and allowing it to flow downward.
- Suitability:
 - Surface irrigation is most suitable for the land with the gentle slopes like terai region of Nepal.
 - This method is suitable for row and field crops.

Surface irrigation is further classified into:

- a. Flooding method: entire land surface is covered with water and allowed to stand on the field.
- b. Contour farming: adapted to hilly areas with steep slopes and quickly falling contours.
- c. Furrow irrigation: common method of irrigation where water is applied in small streams between the rows of the crops, grown on the ridges or furrows.



ii. **Sprinkler irrigation:**

- Sprinkler irrigation is an improvement over conventional surface irrigation method.
- This method simulates natural rainfall to spread water in the form of rain uniformly over the land surface as needed at uniform pattern and at less rate than the infiltration rate of the soil to avoid the surface runoff.
- Suitability:
 - Sprinkler irrigation is highly suitable for sandy soils, rolling and uneven terrain.
 - It is suitable for closely spaced crops like cereals and fodders.



iii. **Drip irrigation:**

- Drip irrigation is the method of applying water directly to the plants through a number of low flow rate outlet called emitter or drippers, placed at short interval along with small tubing.
- The network of drippers supply water and fertilizer to the plant roots at the rate of 1 to 2 liter per hour.
- Suitability:
 - This method of irrigation is highly suitable for arid regions and sloppy terrain where the water is scarce.
 - The method is ideal for row crops, orchard and vineyards.



- Small-Scale Irrigation Project is an irrigation project developed to provide reliable irrigation to a limited command area, managed and operated by local user's communities.
- These projects are designed using simple, low-cost, and locally appropriate technologies to increase agricultural production and improve rural livelihoods.
- In Nepal, small-scale irrigation generally includes:
 - a. Farmer Managed Irrigation Systems (FMIS)
 - b. Small surface irrigation schemes
 - c. Groundwater irrigation (shallow/deep tube wells)
 - d. Lift irrigation schemes
 - e. Pond and rainwater harvesting systems
 - f. Micro-irrigation (drip and sprinkler)
- Scope of small-scale irrigation in Nepal:
 - i. Small scale irrigation creates employment opportunities and increases the income through agricultural productivity.
 - ii. It helps to maintain food security and stable food production throughout the year as small scale irrigation enables multiple cropping and year-round production.
 - iii. Small scale irrigations promote effective utilization of natural resources and local raw materials.
 - iv. It is effective in hills and mountainous region of Nepal where large irrigation projects are difficult to construct.
 - v. Use of Farmer's Managed Irrigation system in small-scale irrigation projects encourage community participation.

3. **Define critical depth of flow in canal. Design a regime canal for a discharge of 15 m³/sec with silt factor=1. Assume a trapezoidal section having side slopes 0.5:1 (H: V)**

Answer:

Critical depth is the depth of flow at which the specific energy is minimum for a given discharge.

At this depth, the Froude number (Fr) = 1, and the flow changes from subcritical to supercritical.

Given,

$$\text{Discharge, } Q = 15 \text{ m}^3/\text{s}$$

$$\text{Silt factor, } f = 1$$

$$\text{Sideslope} = 0.5:1 (H:V)$$

By Lacey's equation,

$$\begin{aligned} P &= 4.75 \sqrt{Q} \\ &= 4.75 \sqrt{15} \\ &= 4.75 (3.873) = 18.40 \text{ m} \end{aligned}$$

According to Regime Velocity equation:

$$V = \sqrt{\frac{2}{5} f d}$$

Since depth d is unknown,

$$Q = AV$$

For trapezoidal section:

$$A = d(B + 0.5d)$$

$$P = B + 2d\sqrt{1 + 0.5^2}$$

$$\sqrt{1 + 0.5^2} = 1.1181$$

Hence,

$$P = B + 2.236d$$

Since $P = 18.40$,

$$B = 18.40 - 2.236d$$

Substituting into area:

$$A = d(18.40 - 2.236d + 0.5d)$$

$$A = 18.40d - 1.736d^2$$

Now,

$$15 = (18.40d - 1.736d^2)\sqrt{0.4d}$$

By trial,

$$\text{For } d = 1.35 \text{ m:}$$

$$B = 18.40 - 2.236(1.35) = 15.38 \text{ m}$$

$$A = 1.35(15.38 + 0.675) = 1.35(15.38 + 0.675) = 21.67 \text{ m}^2$$

$$V = \sqrt{0.4 \times 1.35} = 0.735 \text{ m/s}$$

$$Q = AV$$

$$= 21.67 \times 0.735 = 15.93 \text{ m}^3/\text{s}$$

This is slightly high.

$$\text{Taking } d \approx 1.30 \text{ m,}$$

$$B = 18.40 - 2.236(1.30) = 15.49 \text{ m}$$

$$A = 1.30(15.49 + 0.65) = 20.98 \text{ m}^2$$

$$V = \sqrt{0.4 \times 1.30} = 0.721 \text{ m/s}$$

$$Q = 20.98 \times 0.721 \approx 15.1 \text{ m}^3/\text{s}$$

This is sufficiently close to the required discharge.

4. **Derive Manning's equation for the regime flow of a canal design with consideration of Chezy's equation.**

Answer:

Chezy's equation for uniform (regime) flow in an open channel:

$$V = C\sqrt{RS}$$

Where:

V = Mean velocity of flow (m/s)

C = Chezy's coefficient

R = Hydraulic radius = A/P (m)

S = Bed slope (hydraulic gradient)

Manning related Chezy's coefficient to channel roughness as:

$$C = \frac{1}{n} R^{1/6}$$

Where:

n = Manning's roughness coefficient

Substituting the value of C :

$$V = \left(\frac{1}{n} R^{1/6} \right) \sqrt{RS}$$

Therefore,

$$V = \frac{1}{n} R^{1/6} \times R^{1/2} \times S^{1/2}$$

Adding exponents of R :

$$R^{1/6+1/2} = R^{1/6+3/6} = R^{4/6} = R^{2/3}$$

Hence,

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

This is Manning's velocity equation.

Since,

$$Q = AV$$

Substitute the Manning velocity:

$$Q = \frac{A}{n} R^{2/3} S^{1/2}$$

Where:

Q = Discharge (m³/s)

A = Cross-sectional area (m²)

R = Hydraulic radius (m)

S = Bed slope

n = Manning's roughness coefficient

Velocity equation:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

5. **Write short notes on:**

- i. Hydraulic jump:
- ii. Advantage of canal lining
- iii. Exit gradient
- iv. Cause of failure of weir on permeable foundation
- v. Water measuring devices used in water management

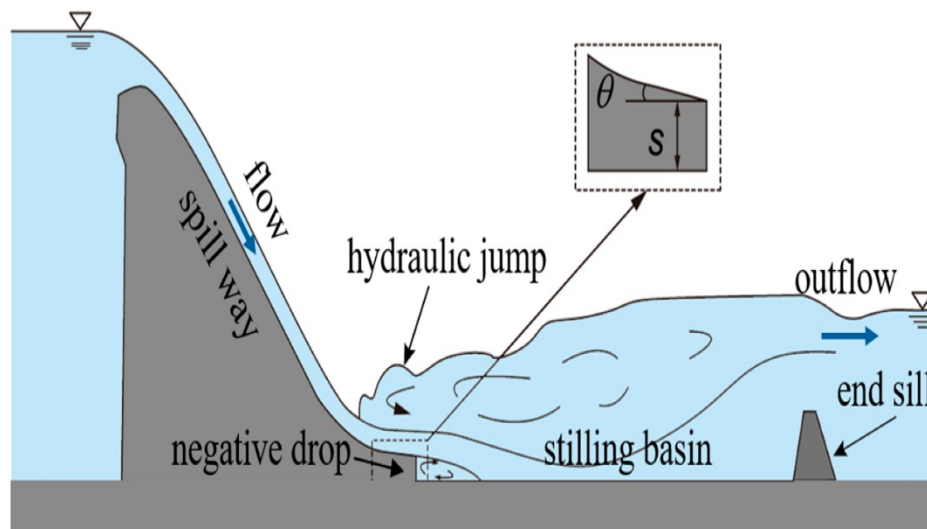
Answer:

i. Hydraulic jump:

- When the depth of flow changes rapidly from a low stage to a high stage, there is abrupt rise of water surface. This local phenomenon is known as hydraulic jump.
- It occurs when the flow changes from supercritical to subcritical.
- Hydraulic jump converts kinetic energy of stream flowing rapidly into potential energy.
- It occurs frequently in a channel below a regulating sluice, at toe of spillway and when the channel bed slope suddenly changes from the steeper to flat slope or where the velocity is high.
- Hydraulic jump forms eddy turbulences in water which causes dissipation of energy.

It is necessary to be formed in hydraulic structures which is explained in the points below:

- i. For the dissipation of excess energy of flowing water over spillway and other hydraulic structure.
- ii. For preventing from possible erosion of downstream floor at structure.
- iii. For increasing the weight of apron floor of hydraulic structure due to increased flow depth.
- iv. For mixing chemicals in water by the help of turbulence created.
- v. For raising the water level in the channel for irrigation.
- vi. For increasing the discharge through a sluice by holding back tail water.



For a stable and well-balanced hydraulic jump, the Froude number (Fr) before the jump should be greater than 1 i.e., the flow must be supercritical.

The table below shows the Froude number for different type of hydraulic jump.

Froude Number (Fr_1)	Type of Hydraulic Jump
$(Fr = 1)$	No hydraulic jump (critical flow)
$(1 < Fr_1 \leq 1.7)$	Undular jump (weak, unstable)

Froude Number ((Fr ₁))	Type of Hydraulic Jump
(1.7 < Fr ₁ ≤ 2.5)	Weak jump
(2.5 < Fr ₁ ≤ 4.5)	Oscillating jump
(4.5 < Fr ₁ ≤ 9)	Stable (steady) hydraulic jump
(Fr ₁ > 9)	Strong jump

A stable and well-balanced hydraulic jump is generally obtained when:

$$4.5 < Fr_1 \leq 9.0$$

ii. Advantages of canal lining:

Canal lining is the method of reducing the seepage loss and improving the canal performance by covering bed and sides of canal by impervious materials such as concrete, mud, steel etc.

The advantages of canal linings are listed as follows:

- Canal linings prevent seepage loss which increases water availability at the tail end of the canal.
- Smooth lining reduces hydraulic resistance allowing higher flow velocity and greater discharge.
- It reduces maintenance cost, minimizing weed growth, silt deposition and bank erosion.
- It also prevents erosion, protecting bank and canal beds from scouring caused by water flow.
- Canal linings also improves irrigation efficiencies with reduced loss of water.
- This method helps to increase canal life by protecting it against weathering, erosion and structural deterioration.

iii. Exit gradient:

The hydraulic gradient of seepage flow at the downstream end of a hydraulic structure where seepage water emerges from beneath the impervious floor is known as exit gradient. It is the rate of loss of hydraulic head per unit length at the point where seepage exits. The exit gradient is a criterion for checking the safety of hydraulic structures against piping and undermining.

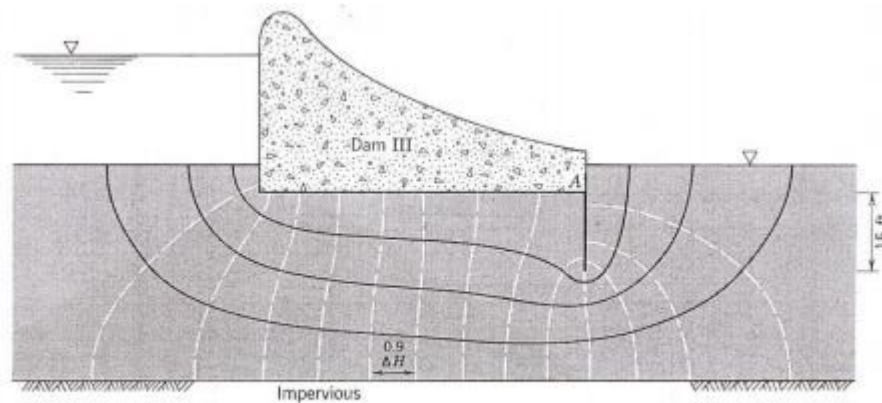


Fig. E18.3 Flow under three dams.

The exit gradient is represented as:

$$G_E = \frac{\Delta h}{\Delta l}$$

where :

$$G_E = \text{Exit gradient}$$

$$\Delta h = \text{Loss of hydraulic head near the exit (m)}$$

$$\Delta l = \text{Length - which the head is lost (m)}$$

According to Khosla's theory, the computed exit gradient must satisfy:

$$G_E \leq G_{Ec}$$

where:

G_E = Calculated exit gradient

G_{Ec} = Critical exit gradient of the foundation soil

If the exit gradient is not balanced, structure fails by quick conditions, piping, undermining or structural failure.

iv. Cause of failure of weir on permeable foundation:

A weir is an impervious barrier constructed across a river or stream to raise the upstream water level, thereby facilitating the diversion of a specific quantity of water into a conveyance system like an irrigation canal or a hydropower intake. The major causes of failure of weir in permeable foundations are:

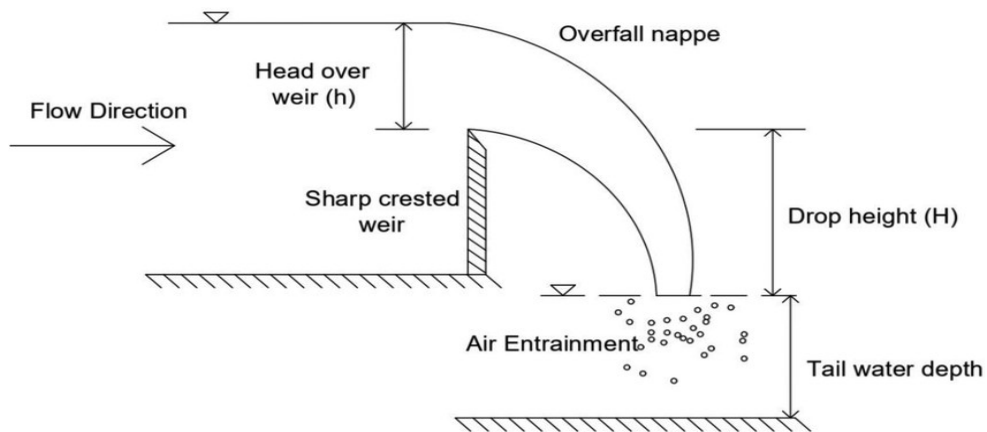
- a. Weir may fail due to formation of underground tunnels (pipes) as seepage water carries soil particles beneath it.
- b. Water seeping beneath the floor exerts upward pressure which may lead to crack, heave or lift on the floor.
- c. Excessive exit gradient at the downstream end causes boiling or quick sand condition which causes erosion, piping and instability.
- d. Differential settlement of permeable foundation of weir produces cracks and structural failure.
- e. Excessive water pressure may also cause weir to fail by overturning or sliding.

v. Water measuring devices used in water management:

Water measuring devices are used to measure the discharge and volume of water in canals and other structures in irrigation, hydropower and water supply systems for efficient water distribution and management. Some of the water measuring devices used in water management are described below:

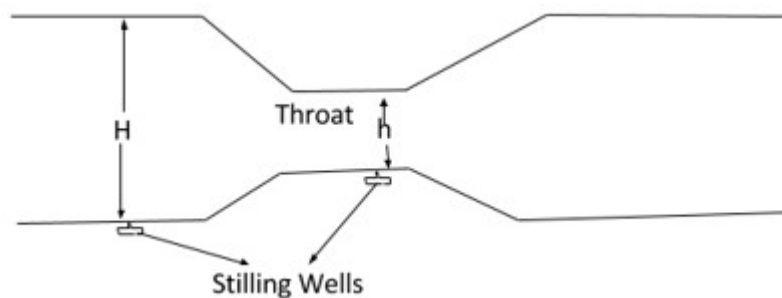
a. Weirs:

- A weir is an impervious barrier constructed across a river or stream to raise the upstream water level, thereby facilitating the diversion of a specific quantity of water into a conveyance system like an irrigation canal or a hydropower intake.
- They are suitable for small to medium discharges.
- There are different types of weirs. i.e., rectangular weir, trapezoidal weir, triangular weirs and so on.



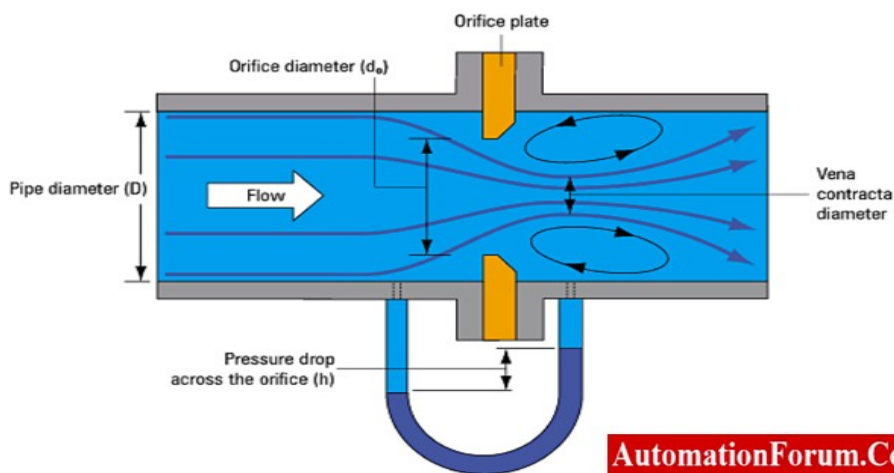
b. Flumes:

- Flumes are suitable for measuring discharges in irrigation canals with minimal head loss.
- Types: Parshall flumes, Cut-throat flume, venturi flume etc.



c. Orifices:

- Orifices are used to measure discharge from reservoir, tanks and canals.
- It is a type of flow meter that measures the flow rate of fluids (liquids or gases) based on the principle of differential pressure.



d. Current meters:

- Measure the velocity of flowing water.
- Discharge is calculated using:

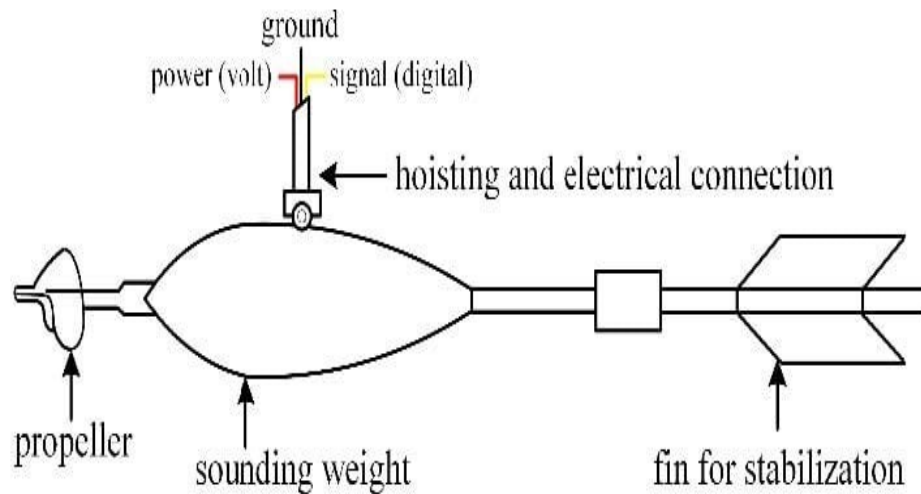
$$Q = A \times V$$

where:

Q = discharge (m^3/s)

A = cross-sectional area (m^2)

V = average velocity (m/s)



6. **How serious is the problem of water logging in irrigated area in Nepal? State the various drainage systems required in removing water logging in the irrigation fields?**

Answer:

Water logging is the condition of saturation of soil such that air phase is restricted and anaerobic conditions prevails. Waterlogging is often accompanied by soil salinity as waterlogged soils prevent leaching of the salts imported by the irrigation water. Due to water logging, groundwater table rises close to the ground surface (generally within 1.5–2.0 m), causing the root zone to remain saturated and reducing soil aeration.

- Water logging is an increasing problem in flat terrains of terai region and low-lying pocket lands in Nepal.
- It is a moderately serious, localized, but recurring seasonal problem.
- It is primarily driven not only by over-irrigation but also by heavy monsoon rainfall combined with unmanaged infrastructure.
- Water-logging is caused by intensive irrigation, seepage from canals/ reservoirs, seepage from adjoining lands, lack of adequate drainage systems, monsoon rainfall and so on.
- Water logging has caused hamper in nitrification which reduces the crop yield and also leads to rise of salt in surface layer and overgrowth of weeds.

The various drainage system required in removing water logging in the irrigation fields are given below:

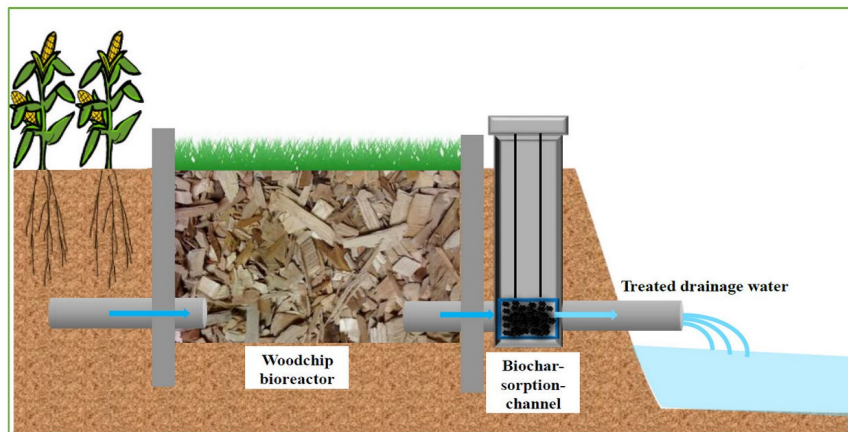
i. Surface drainage system:

- The removal of excess rainwater falling on the fields or excess irrigation water applied to the fields by constructing open ditches. Field drains and other structures is known as surface drainage system.
- The land is loped toward the ditches or drains making excess flow of water to remove from the drains.
- Surface drains can be divided into surface drain and deep surface drains.



ii. Subsurface drainage system:

- The removal of excess rainwater by using buried perforated pipes or tile drains to lower the groundwater table is known as subsurface drainage system.
- It is most effective for controlling water logging in agricultural fields.



iii. Interceptor drains:

- These drains are constructed along the canals or hill slope to intercept seepage before it reaches agricultural land.

iv. Bio-drainage:

- Plantation of deeply rooted trees, highly water absorbing plants to lower the groundwater table naturally.

v. Open drainage system:

- It consists of open ditches that carry water to nearby rivers or canals.
- It is simple and economical for large agricultural areas.

7. What is river training works? Describe briefly the different types of river training works and its usefulness in the various conditions.

Answer:

River training works are the structural engineering measures constructed along rivers to control the flow, stabilize the channel, and prevent bank erosion. They are designed to guide water within a specific path, safely pass floodwaters, and protect nearby infrastructure from shifting currents.

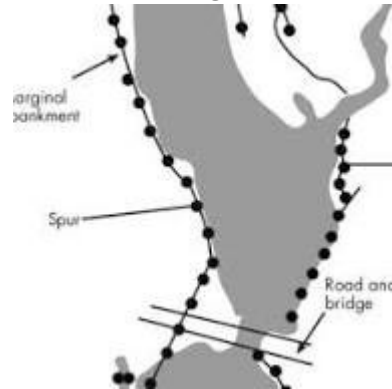
In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting. Several projects are carried out to construct massive embankments and guide bunds to protect settlements and agricultural lands near Koshi river and its tributaries. Floods are reported to cause roughly 100

to 175 deaths per year on average and water-induced disasters reports about NPR 762 million per year in combined annual losses.

There are different methods of river training works. Some of them are described below:

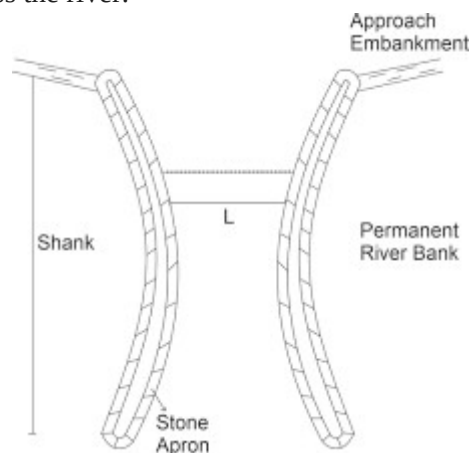
i. Marginal embankments/ levees:

- Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
- They are usually placed at some distance away from the riverbank so that area of the flow is increased.
- Generally required for meandering rivers.



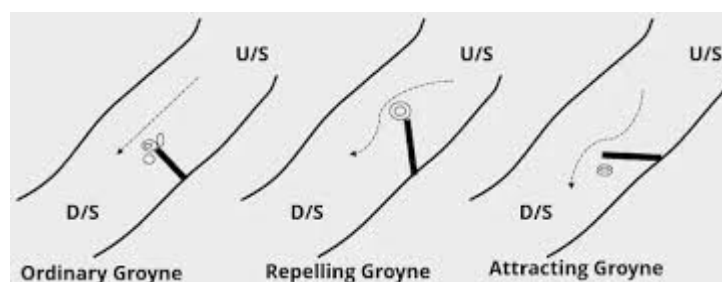
ii. Guide banks:

- Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change the through the waterway at bridge site.
- The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.



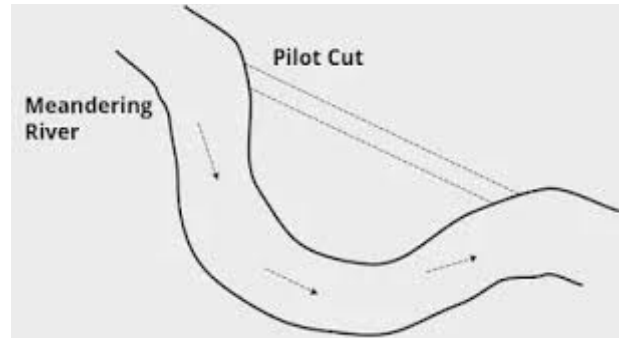
iii. Groynes or spurs:

- Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
- They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



iv. Artificial cutoff:

- An artificial cutoff is developed in a meandering river to divert the flow along a straight course.
- They are constructed for training of river when its mender goes on increasing and endanger some valuable land or property on the banks.



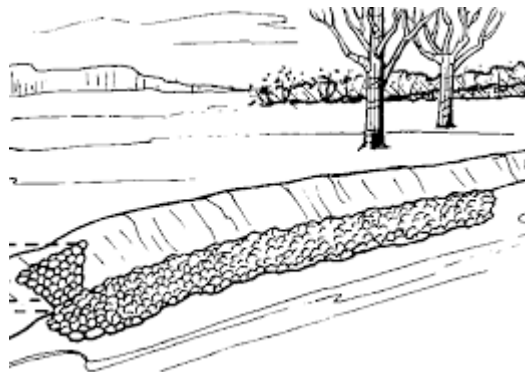
v. Bank protection and Pitched banks:

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

Bank protection work can be classified as direct or indirect.

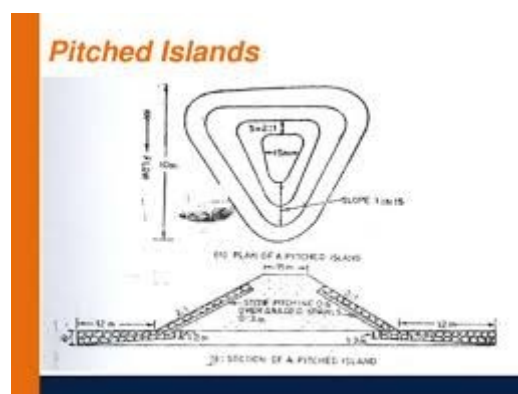
Direct protection works include work done on the bank itself such as vegetation cover, pitching

Indirect protection includes work constructed not directly on the banks but in front of them.



vi. Pitched islands:

- Pitch Island is an artificially created island in the river bed, either made up of masonry or stone pitched on all sides.
- The main purpose of pitched island is to train the river to have an axial flow and improve the channel for navigation purpose.



vii. Bandalling:

- Bandalling is a method of river training works which is used to confine the low flow water of a shallow river in a single channel for maintaining the required depth for navigation.
- Bandalling started after the flood season is over and when the water level begins to fall.



8. **What effect you foresee on hydrological cycle due to climate change? Suggest the ways and means in mitigating climate change effects on irrigation projects.**

Answer:

Climate change is a critical challenge globally and Nepal is not an exception. It ranks 10th globally among the countries with high risk of climate change. The effects that can be foreseen on the hydrological cycle due to climate change are described in detailed below.

i. **Glacial Lake Outburst Floods (GLOFs):**

Increased temperatures cause ice caps and glaciers to melt at an accelerated rate, leading to the formation of unstable glacial lakes and a higher risk of GLOFs, which can devastate downstream ecosystems and infrastructure.

ii. **Seasonal and Rainfall Imbalance:**

Patterns are shifting toward intensified and untimely rainfall, which increases surface runoff and severe flooding during the monsoon, while simultaneously leading to prolonged dry winters and droughts.

iii. **Drying of Water Resources:**

Increased earth temperatures and insufficient rainfall have resulted in the drying up of traditional natural water resources, such as springs (dhunge dharas) and streams.

iv. **Increased Sedimentation:**

Intensified rainfall triggers more landslides and erosion in steep terrains, significantly increasing the sediment load in rivers; this raises riverbeds, reduces channel capacity, and heightens flood risks.

The ways of mitigating climate change effects on irrigation projects are given below:

i. **Catchment Ponds and Reservoirs:**

Constructing catchment ponds can trap excess surface runoff for groundwater recharge. Installing multiple elevated reservoirs allows for storing excess monsoon water to be utilized during the water-scarce dry season.

ii. **Sediment Management:**

Intakes should be equipped with sediment filtration systems to prevent debris and silt from entering irrigation canals, which otherwise reduces system efficiency.

iii. **Managed Aquifer Recharge (MAR):**

This involves directing stormwater and treated wastewater into the ground to recharge depleted aquifers.

iv. **Rainwater harvesting:**

Collecting and treating runoff from roofs and open spaces can supplement irrigation needs and reduce the over-encroachment of natural water bodies.

v. **Bio-engineering:**

Utilizing vegetative systems (like turfing or fascines) alongside civil structures provides flexible, cost-effective slope stabilization to protect irrigation canals from landslides.

vi. **Environmental Assessments:**

Mandatory Environmental Impact Assessments (EIA) or Initial Environmental Examinations (IEE) should be conducted to ensure projects are environmentally sustainable and resilient to climate-induced geological instability.

9. What do you mean by peak flow? What is the importance of having a correct peak flow estimation? State the different methods of peak flow estimation.

Answer:

The maximum rate of flow of water passing through a particular point in a river, stream, canal, or watershed during a storm or flood event is known as peak flow. It is usually expressed in m³/s (cumecs). Peak flow is also known as peak discharge.

The importance of having a correct peak flow estimation are as follows:

- i. Correct peak flow estimation is used for correct designing dams, spillways, weirs, barrages, culverts, bridges, and canals.
- ii. It also helps to predict flood magnitude and design of flood protection structures.
- iii. It also ensures the adequate design of stormwater drainage and urban drainage systems.
- iv. Peak flow estimation prevents overtopping, failure and damage to hydraulic structures.
- v. It helps to determine the reservoir storage and spillway discharge capacity.

The different methods of peak flow estimation are as follows:

i. Rational method:

Rational method is suitable for the catchment with limited catchment area)

$$Q = CIA/360$$

where:

Q = Peak runoff (m³/s)

C = Runoff coefficient

I = Rainfall intensity (mm/hr)

A = Catchment area (ha)

ii. Empirical method:

Empirical formula is suitable for the catchment with large catchment area.

Dicken's Formula:

$$Q = C A^{3/4}$$

where:

Q = Peak runoff (m³/s)

A = Catchment area (km²)

C = Dicken's coefficient (depends on region)

Ryve's Formula:

$$Q = C A^{2/3}$$

where:

Q = Peak runoff (m³/s)

A = Catchment area (km^2)

C = Ryve's coefficient

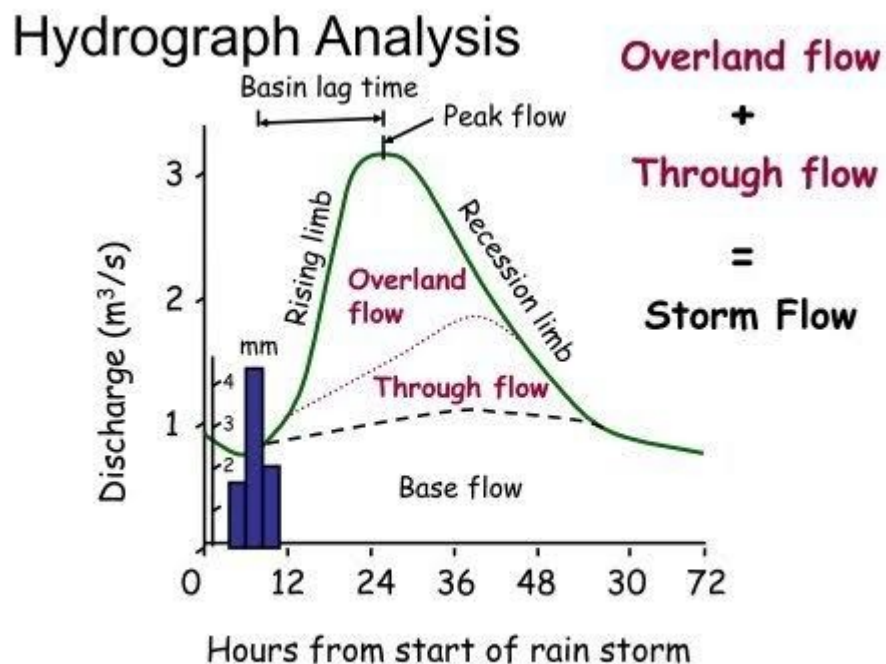
iii. **Unit hydrograph method:**

Unit hydrograph is the direct run of hydrograph resulting from unit depth (1cm) of excess rainfall occurring at the uniform rate for the specific duration of D hours.

Unit hydrograph is the unit pulse response function of linear hydrologic system.

The basic assumptions of the unit hydrograph theory in flood forecasting in rivers of Nepal are listed below:

- The excess rainfall has constant intensity one by $\frac{1}{D}$ cm/hr for effective storm duration of D hours.
- The excess rainfall is uniformly distributed over the entire catchment area.
- The base time of direct runoff hydrograph (DRH) resulting from excess rainfall of given duration is constant.



10. Describe the different method of recharging ground water table preserving ecological balance in nature of south Asian regions.

Answer:

The process of refilling the depleted water level of underground aquifers by rainwater or surface water that infiltrates into the soil and replenishes them is known as groundwater recharge. Artificial and natural recharge methods are used to maintain groundwater levels which helps to ensure sustainable water supply, and preservation of ecological balance.

The different methods of recharging ground water table preserving ecological balance are described below:

i. **Managed Aquifer Recharge (MAR):**

- Managed Aquifer Recharge is a technical water management strategy where stormwater, surface water, or even treated wastewater is intentionally directed into the ground.
- This technique is designed specifically to recharge depleted aquifers and enhance natural infiltration processes.

ii. **Rainwater harvesting and recharge structures:**

- Rainwater harvesting involves collecting precipitation runoff from **roofs, yards, and open spaces**.

- While this water can be stored for immediate household use, it is also frequently directed into specialized **recharge structures** to replenish groundwater levels.
- This is particularly effective in areas where traditional water sources are drying up.

iii. Catchments and Recharge Ponds:

- The construction of **catchment ponds** is a common method used to trap excess surface runoff, especially during the monsoon season.
- These ponds hold water long enough to facilitate its gradual infiltration into the underlying strata, utilizing what would otherwise be wasted floodwater to recharge local aquifers.

iv. Watershed protection and afforestation:

- **Forest Conservation:** Forests and vegetative covers reduce the velocity of surface runoff, which allows more time for water to **infiltrate and percolate** into the soil.
- **Vegetative Systems:** Deep-rooted trees and plants act as natural anchorage, strengthening soil mass and facilitating the movement of water into the groundwater table.

v. Urban planning:

- **Park Construction:** Establishing parks and green spaces provides natural permeable areas for recharge.
- **Reducing Concrete Coverage:** Minimizing non-permeable surfaces facilitates the conservation and natural recharge of groundwater levels.

vi. Greywater management:

- In water-scarce rural hilly regions, greywater management serves as a low-cost recharge option.
- Domestic wastewater from bathrooms and sinks (excluding organic waste) can be collected, treated, and filtered back into the soil.
- This process not only prevents water stagnation but also helps replenish shallow local aquifers.

